

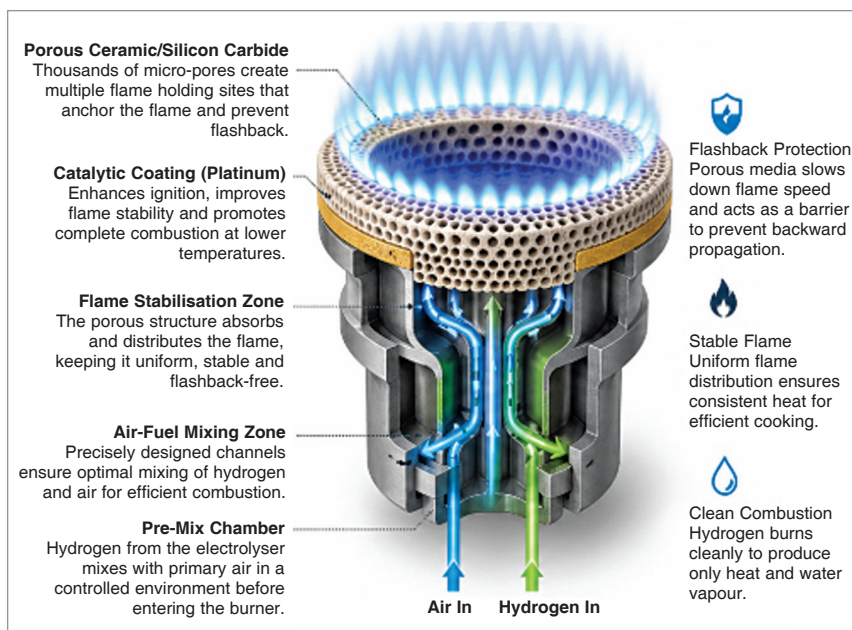


Fig. 2: A burner cross-section graphic showing porous ceramic and flame stabilisation

electrolysers have enabled early-stage testing of hydrogen stoves.

Hydrogen can be produced on demand using electricity and water, eliminating the need for cylinders or dedicated storage infrastructure. According to prototype estimates, 1kWh electricity and approximately 100 millilitres of water may provide enough energy for several hours of cooking. Hydrogen has demonstrated significant versatility as an energy carrier and cooking fuel; however, the key question for large-scale deployment is whether efficiency or electricity availability will prove to be the greater constraint.

Although hydrogen stoves remain expensive (approximately ₹150,000), they are already commercially available and in use. The key distinction lies in the underlying fuel model:

- **LPG.** Centralised production and distribution
- **Hydrogen.** Decentralised, site-level production and immediate consumption

Like LPG, the long-term viability of hydrogen cooking will ultimately depend on affordability. Electricity pricing, energy availability, and cost competitiveness relative to LPG and induction cooking will be

decisive factors. If costs remain high, consumers are likely to favour cheaper alternatives, even if those options are less environmentally sustainable.

The adoption of hydrogen cooking in India is likely to follow a trajectory similar to that of previous technological transitions, progressing gradually as economic viability allows. As with earlier innovations, adoption would begin among early adopters before expanding to broader market segments. In the short term (0-5 years), uptake is likely to remain limited to premium urban households, hotels, restaurants, and smart-city initiatives. In the medium term (5-10 years), adoption could expand

into semi-urban areas as costs decline and integration with rooftop solar systems improves. Beyond ten years, hydrogen cooking may begin to gain traction in rural areas as well.

A cleaner flame, but not a complete solution yet

Fossil fuels such as coal, kerosene, and firewood, as well as poor-quality LPG, emit harmful pollutants that contribute significantly to indoor air pollution. Hydrogen, by contrast, primarily produces water vapour during combustion, reducing many of the health risks associated with conventional cooking fuels, particularly when the hydrogen itself is produced using renewable energy.

However, if hydrogen is generated using electricity from fossil-fuel-based power plants, emissions are effectively shifted upstream rather than eliminated. Consequently, the method used to produce hydrogen is a critical factor in determining its environmental credentials.

In principle, hydrogen is a viable cooking fuel. Nevertheless, several important challenges remain:

- The initial cost remains high, ranging from ₹50,000 to ₹150,000
- The system requires a steady supply of electricity
- Hydrogen fuel cell systems have a relatively shorter lifespan, including components like PEM electrolyzers and catalysts
- Hydrogen's flammability raises safety concerns

Comparison with existing technologies

Whereas conventional gas cooking appliances rely on centralised fuel production and distribution, hydrogen-powered systems emphasise decentralisation by generating fuel on-site within the appliance.

While hydrogen stoves offer advantages in terms of low emissions and decentralised fuel generation, they currently lag behind established alternatives in cost and ecosystem maturity. LPG remains dominant because of its affordability and

Parameter	Hydrogen Stove	LPG	Induction
Emissions	Zero (in use)	CO ₂	Zero
Efficiency	~75-80%	~60%	~85-90%
Fuel Source	Electricity	Fossil	Electricity
Infrastructure	None	Cylinder	Grid
Cost (initial)	High	Low	Medium

extensive distribution network, while induction cooking leads in efficiency and convenience where reliable electricity is available. Hydrogen's distinctive advantage lies in its potential to bridge fuel-based and electric cooking paradigms, combining clean combustion with on-demand fuel generation.